Ira A. Fulton Schools of Engineering
Arizona State University

# Fall 2026/Spring 2027 Interdisciplinary Capstone Project Request Form

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|---|---|---|---|
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## Project Title

**Automated Delivery of Material/Parts to Process Tools Including Machine Integration for Full Automated Replenishment**

## Project Summary (See page three for instructions)

- As part of the movement to increased automated operations in the backend assembly processes for semiconductors, there is a need to automate typical human operations for replenishment and delivery of items to process tools.
- To enable this the robotic systems, controls architecture and the mechanical aspects of the delivery need to be developed and standardized to allow for automated delivery.
- The concept is to utilize Autonomous Mobile Robots with integrated secondary robotics to perform these tasks.
- This project scope is to define the operational model, review and develop concepts around the delivery methods, define the controls aspects for managing the delivery and design delivery points and container concepts that can be standardized for this function.

- Key Concepts:
  - Definition of material scope and proposed ‘carrier’ designs.
  - Integration aspects for mechanical delivery and the enabling to the autonomous robots.
  - Specific standardization of the payloads and delivery points for the Process tools.

| Engineering Disciplines Required | None | Low | Med | Hi | Engineering Disciplines Required | None | Low | Med | Hi |
|---|---|---|---|---|---|---|---|---|---|
| Mechanical | ☐ | ☐ | ☒ | ☐ | Aerospace | ☒ | ☐ | ☐ | ☐ |
| Electrical | ☐ | ☐ | ☒ | ☐ | Computer Science | ☐ | ☐ | ☒ | ☐ |
| Industrial | ☐ | ☐ | ☒ | ☐ | Other | ☐ | ☐ | ☐ | ☐ |

| Specialized Skills Required | Low | Med | Hi |
|---|---|---|---|
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |
| | ☐ | ☐ | ☐ |

US Citizen ONLY ☐

IP Agreement or NDA Required ☒

Travel Required ☐

**Additional Comments:**

There will be opportunity for prototyping and concept demonstration for the containers and the delivery concepts. There is a possibility for work directly with AMRs to demonstrate the capabilities, but that will need to be finalized.

The expectations are to define the overall ecosystem with targeted demonstrations for the management and delivery of the materials.